

Unsupervised Concept Discovery and Evaluation for Medical Vision-Language Models

Riccardo Marconi
Politecnico di Torino
Turin, Italy

Davide Candela
Politecnico di Torino
Turin, Italy

Emmanuel Messina
Politecnico di Torino
Turin, Italy

ABSTRACT

We investigate unsupervised concept discovery in medical vision-language models through a chest X-ray adaptation of MedConcept. A sparse autoencoder decomposes frozen BiomedCLIP embeddings into 1,024 latent features, named using 998 phrases associated with 100 clinical concepts. We train on 7,500 NIH images [4], then extract concepts from Open-I images [14] and evaluate them against the corresponding reports using a frozen MedGemma evaluator. Across 500 images, mean per-image scores are 7.21% Aligned, 22.49% Unaligned, and 70.30% Uncertain. We further investigate alignment using hierarchical clustering, revealing substantial variation across clinical subdomains: the proportion of aligned concepts ranges up to 49.23% across clusters. We complement this analysis with a blinded audit of 25 concept-report pairs. MedGemma agrees with the three authors’ majority verdict on 56% of audited pairs, compared with 68-76% for the commercial evaluators tested. These results demonstrate a working concept-discovery pipeline while exposing the limitations of report-based evaluation: sparse reconstruction, textual agreement, and clinical validity are distinct properties.

1 INTRODUCTION

Medical vision-language models (VLMs) encode images and text in a shared representation and have achieved strong performance across clinical tasks. However, their high-dimensional, polysemantic latent representations make the clinical meaning of individual dimensions difficult to establish, limiting interpretability and trustworthiness in safety-critical applications. Concept-based explanations aim to express this representation through recognizable anatomy and findings. Following MedConcept [6], we study whether sparse autoencoder (SAE) features extracted without diagnostic supervision can be named and assessed against radiology reports.

We adapt the original 3D abdominal CT framework to 2D chest radiographs using BiomedCLIP [15]. Our contribution is an implemented pipeline with phrase-level semantic grounding, an analysis of concept groups at several clustering resolutions, and an independent, multi-evaluator audit of the language-model judge that also surfaces the pipeline’s dependence on external resources. The objective is to assess what the available evidence supports about interpretability, rather than to develop a diagnostic classifier. We find that report-based alignment is low on average but highly uneven across clinical subdomains, and that the choice of judge itself, not just the discovered concepts, is a major source of disagreement in how that alignment is measured.

2 RELATED WORK

From predefined concepts to discovery. TCAV measures prediction sensitivity along directions learned from examples of user-defined concepts [7]. Concept Bottleneck Models instead predict

annotated concepts as an intermediate representation and permit interventions before the final prediction [8]. Both offer semantic interfaces, but rely on a supplied concept specification. Sparse dictionary learning removes this requirement by decomposing polysemantic activations into more monosemantic features without predefined concepts [3], and Olah et al. [9] frame the resulting units and their connections as interpretable circuits rather than opaque activations. Building on this idea, Discover-then-Name applies SAE discovery before naming concepts and training downstream concept bottlenecks [11], while SpLiCE instead decomposes CLIP embeddings directly into sparse combinations of existing textual concepts, without training a new representation [1].

VLMs and medical grounding. Pach et al. [10] evaluate SAE features in VLMs using a human-study benchmark and interventions, going beyond reconstruction alone. Closer to our clinical setting, Gong et al. [5] dissect neuron activations in medical image diagnostic models to build interpretable, concept-level explanations directly from opaque networks, motivating a neuron-to-concept alignment strategy for our chest radiograph setting rather than a supervised diagnostic head. MedConcept [6] combines sparse decomposition of Merlin CT embeddings, an ontology-derived vocabulary, cosine-based naming, and frozen MedGemma [12] judgments of concept-report consistency. We retain this separation between discovery and evaluation, while changing the image domain, backbone, vocabulary, and activation selection.

Evaluating explanations. Different evaluation methods capture distinct aspects of explanation quality: reconstruction measures how well embeddings are preserved, human inspection assesses semantic coherence, interventions test effects on model behavior, and report entailment assesses textual support. These approaches provide complementary evidence. Although LLM-based evaluation scales more readily than manual review, comparative judging studies have demonstrated susceptibility to position bias [13]. We therefore use label permutation and independent validation to improve evaluation reliability, while recognizing that neither guarantees error-free judgments.

3 RESEARCH GAPS

Four gaps guide our analysis. First, an SAE’s sparsity and reconstruction fidelity do not guarantee that a nearest text label correctly describes its feature: vocabulary coverage and the image-text geometry constrain naming. Second, the pipeline depends on external resources at every stage that turns raw activations into a clinical claim: an ontology-derived vocabulary, a generative model for phrase expansion, and one or more large language models as semantic judges. Biases or errors in any of these components can be mistaken for properties of the discovered representation itself. Third, a report is selective evidence. An unmentioned concept may

be visible, absent, or incorrectly named, and an external judge can additionally confuse omission with contradiction. Fourth, independently assigned labels omit relationships between concepts. We examine whether semantic clustering produces useful groups, while recognizing that embedding similarity does not establish an anatomical hierarchy or causal relation. Together, these gaps point to the broader difficulty of assessing interpretability in a scalable and reproducible way: our human and multi-model audit addresses evaluator reliability specifically.

4 METHODOLOGY

4.1 Datasets and feature extraction

We use two chest X-ray datasets with distinct roles: the NIH Chest X-ray 10k subset [4] for learning the sparse representation, and Open-I [14] for evaluating the extracted concepts against radiology reports.

The NIH subset contains 10,000 images. We retain its provided split, using the training images to fit the sparse autoencoder (SAE) and the images in the test split as a validation set for early stopping and model selection. The SAE learns exclusively from image embeddings, without report text or disease labels.

Open-I provides chest X-ray images paired with radiology reports, allowing us to assess whether the concepts extracted from an image are supported by the corresponding clinical text. We use the frontal image from each record, excluding records with missing frontal images. For each image, we construct a reference report by concatenating the available *Findings* and *Impression* sections, which describe the radiological observations and their diagnostic interpretation. This preprocessing yields 3,666 image-report pairs, forming the pool from which the semantic evaluation samples are drawn. Open-I images and reports are not used to train the SAE.

We use BiomedCLIP’s frozen ViT-B/16 [15] image encoder for images from both datasets to produce the visual embeddings used for SAE training and inference.

4.2 Concept dictionary construction

To assign interpretable names to the features learned by the SAE, we construct a dictionary of medical concepts. We start from Open-I’s *Problems* metadata, which contains medical terms associated with each record. After normalizing formatting and removing duplicates, we obtain 116 distinct candidate terms. We exclude 15 device, procedure, and technical terms, map broad terms to chest X-ray terminology (e.g., “opacity” to “lung opacity”), and merge any resulting duplicates. This yields 100 parent concepts covering anatomical structures and pathological findings.

We then query the Unified Medical Language System (UMLS) [2] for an exact match to each concept. For matched concepts, we retrieve a Concept Unique Identifier (CUI) and available English synonyms from SNOMED CT and MeSH, two medical terminology resources accessible through UMLS. This step links the vocabulary to standardized terminology while preserving alternative names for the same concept. We obtain a CUI for 90 concepts; the remaining ten retain their base term as a fallback.

To complement these terminology entries with radiology-style descriptions, we use Gemini 3.1 Flash-Lite to generate five short

phrases per concept, providing the base term and available synonyms as input. The prompt requests descriptions of visible features consistent with the concept, including normal anatomy. We check that phrases are unique within each concept and long enough to be semantically meaningful.

Finally, we encode each dictionary entry independently using BiomedCLIP’s PubMedBERT text encoder [15] and L2-normalize its embedding, which lies in the same latent space as the image features. The resulting matrix, $T \in \mathbb{R}^{998 \times 512}$, contains 498 base-term and synonym entries and 500 generated phrases, with each row linked to its parent concept.

4.3 Sparse Autoencoder and concept assignment

We train an SAE to represent each image embedding through a sparse combination of learned features. The input dimension is $d = 512$ and the hidden dimension is $m = 1,024$, giving an overcomplete factor of $m/d = 2$. For an image embedding $x \in \mathbb{R}^d$, the encoder produces non-negative activations $z \in \mathbb{R}^m$, from which the decoder reconstructs the input:

$$z = \text{ReLU}(W_e(x - b_d) + b_e), \quad \hat{x} = W_d z + b_d, \quad (1)$$

Here, W_e and W_d are the encoder and decoder weights, and b_e and b_d are learned biases. Training combines reconstruction error with an L_1 penalty that encourages sparse activations:

$$\mathcal{L} = \frac{1}{512B} \sum_{i=1}^B \|x_i - \hat{x}_i\|_2^2 + \frac{\lambda}{B} \sum_{i=1}^B \|z_i\|_1, \quad (2)$$

where $B = 256$ is the batch size and λ controls the sparsity penalty. We train with Adam and use validation loss for early stopping and model selection.

After SAE training, each decoder column represents a learned feature direction in the image embedding space. We assign a concept phrase to each feature by comparing its decoder column with the dictionary embeddings. This naming step does not affect the learned weights. We compute the similarities as

$$S = T \tilde{W}_d \in \mathbb{R}^{998 \times 1024}, \quad (3)$$

where $\tilde{W}_d$ contains the normalized decoder columns. Since the rows of T also have unit length, S_{pj} is the cosine similarity between phrase p and feature j . Then, each feature receives the phrase with the highest similarity in its column of S . Features are matched independently, allowing multiple features to receive the same phrase label.

Once the learned features have been assigned phrase labels, we use the trained SAE to extract concepts from Open-I images. For each image, we feed its BiomedCLIP embedding into the SAE and select the active neurons whose cumulative activation exceeds 90% of the total positive activation, starting with the largest values. Let $z_{(1)} \geq \dots \geq z_{(M)} > 0$ denote the M positive activations in descending order. The number of selected features is

$$k^* = \min \left\{ k \in \{1, \dots, M\} : \frac{\sum_{j=1}^k z_{(j)}}{\sum_{j=1}^M z_{(j)}} > 0.90 \right\}. \quad (4)$$

This allows the number of selected features to vary across images. We collect the labels of the first k^* features and remove duplicate phrases to obtain the predicted concept set. No minimum cosine-similarity threshold is applied.

4.4 Semantic evaluation and validation

To assess whether the concepts extracted from each Open-I image are supported by its radiology report, we use a frozen MedGemma 1.5 4B [12] as an automated evaluator. For each predicted concept, the model receives its phrase label and the corresponding report, without the image. The prompt provides definitions and examples for three verdicts: *Aligned* when the report explicitly supports the concept; *Unaligned* when it explicitly negates or contradicts the concept; and *Uncertain* when the concept is unmentioned, ambiguous, or tentative.

We map the verdict classes to single-token answers A, B, and C, rotating the class-letter mapping across three forward passes to reduce letter bias. In each pass, we apply softmax to letters’ next-token scores and map the probabilities back to the classes. The class with the highest average probability across passes becomes the verdict. For the overall evaluation, we average the per-image proportions of concepts assigned to each verdict, giving each image equal weight.

To examine alignment across broader concept groups, we apply agglomerative clustering with cosine distance and average linkage to the parent concepts, each represented by the L2-normalized mean of its phrase embeddings. We assess cluster separation using silhouette scores for $k = 3, \dots, 15$. For $k \in \{3, 5, 8, 10\}$, we aggregate the existing MedGemma verdicts by mapping each predicted phrase to its parent concept’s cluster, without querying the evaluator again. Within each cluster, verdict percentages are calculated over phrase-image pairs.

Finally, we assess evaluator reliability on 25 concept-report pairs, sampled by MedGemma’s original verdict: 8 Aligned, 8 Unaligned, and 9 Uncertain. The three authors independently label these pairs with MedGemma’s verdicts hidden, and majority vote defines the reference for measuring evaluator agreement. Three commercial language models also classify the same pairs using both file attachments and pasted text, allowing us to examine sensitivity to input format. This small, stratified audit measures agreement on report interpretation without expert review of the X-ray images.

5 EXPERIMENTS AND ANALYSIS

In this section, we evaluate our concept discovery framework across three core dimensions: (1) *Sparse Autoencoder (SAE) reconstruction quality and sparsity*, (2) *semantic alignment between visual features and clinical radiology reports*, and (3) *macro-structural concept clustering*. Finally, we present an *evaluator audit* comparing human consensus with multi-LLM benchmarks.

5.1 Sparse Reconstruction and Grounding

We trained multiple SAEs on BiomedCLIP embeddings from 7,500 NIH chest X-rays, using 2,500 additional images for validation and varying the sparsity regularization coefficient λ . The results presented here correspond to $\lambda = 5 \times 10^{-5}$. We used Adam with an initial learning rate of 10^{-3} , halving it when validation loss plateaued with a scheduler patience of five epochs. Decoder columns were normalized to unit length after each update. Training was limited to 1,000 epochs, with an early-stopping patience of 25 epochs. For the reported configuration, training stopped at epoch 413, and the checkpoint from epoch 388 was selected because it achieved

the lowest validation loss. On 256 validation embeddings, the selected SAE achieves Mean Squared Error (MSE) 3.32×10^{-5} and mean cosine similarity 0.9916 between original vectors x and reconstructed vectors $\hat{x}$. The activation sparsity reaches average L_1 norm $\|z\|_1 \approx 0.87 - 0.96$, with average L_0 sparsity of 44.0 active units out of 1,024 per image ($\sim 4.3\%$). These metrics confirm near-lossless visual reconstruction under a strict sparse activation regime. However, high reconstruction fidelity is a necessary condition for feature decomposition but does not guarantee semantic monosemanticity [3].

Following the concept assignment procedure described in Section 4.3, we label the SAE features using the concept dictionary. Retaining features accounting for over 90% of the positive activation yields approximately 26 unique concept phrases per image, on average, in the evaluated Open-I subsets.

5.2 Report Alignment and Subset Analysis

Semantic validation measures whether extracted visual concepts are supported by written radiology reports. We employ frozen MedGemma 1.5 4B as an automated judge. To avoid generative latency and text parsing errors, MedGemma evaluates each concept-report pair deterministically via next-token logit probabilities over tokens A, B, C across three label-permuted passes (options *Aligned*, *Unaligned*, and *Uncertain*).

Table 1 reports verdict percentages across Open-I evaluation subsets ($N \in \{10, 50, 100, 500\}$). At $N = 500$, 12,996 phrase-image pairs were evaluated (25.99 unique phrases/image).

Table 1: Mean per-image verdict percentages across Open-I subsets (N denotes the number of images).

N	Phrases/img	Aligned (%)	Unaligned (%)	Uncertain (%)
10	26.20	7.25	23.17	69.57
50	26.22	8.40	22.88	68.72
100	26.23	7.64	23.37	68.99
500	25.99	7.21	22.49	70.30

Across evaluation subsets, verdict distributions remain broadly stable: approximately 70% Uncertain, 22.5% Unaligned, and 7.2% Aligned. The predominance of Uncertain verdicts is consistent with reporting omission bias: radiologists prioritize clinically relevant findings and may omit normal anatomy or other visible features. Consequently, a concept may be present in an image without receiving explicit textual support in its report. The 22.5% Unaligned rate reflects predicted concepts that the evaluator identifies as contradicted or explicitly negated by the corresponding reports, suggesting potential false-positive predictions. These results highlight the distinction between a concept’s visual presence and its documentation in clinical text.

5.3 Hierarchical Concept Clustering

To analyze how alignment varies across clinical subdomains, we perform agglomerative hierarchical clustering on the 100 parent concept embeddings using cosine distance and average linkage. Silhouette analysis across $k \in [3, 15]$ reveals a peak score of 0.274 at $k = 3$. Among the four resolutions for which we aggregate

verdicts ($k \in \{3, 5, 8, 10\}$), $k = 10$ scores second-highest (0.263, close to the $k = 3$ peak), while $k = 8$ scores considerably lower (0.184). We therefore adopt $k = 10$ as our reference resolution: it retains near-peak cluster separation while offering finer granularity than $k = 3$. Clusters are referred to only by the numeric identifier assigned by the clustering algorithm; we do not assign them descriptive names, since we lack the clinical expertise to validate such labels ourselves.

Table 2 reports verdict percentages for all ten clusters at $k = 10$ ($N = 500$). Most clusters sit close to the overall average (Table 1), but three stand out: Cluster 9 reaches 49.23% Aligned, the highest of any group; Cluster 5 reaches only 0.15%, the lowest; and Cluster 1 reaches 42.02% Unaligned, the highest disagreement rate observed.

These extreme alignment rates remain unchanged across all evaluated resolutions, $k \in \{3, 5, 8, 10\}$, with both concept groups forming separate clusters even at $k = 3$. This consistency suggests that the observed differences in report-based agreement are robust to clustering resolution and can be identified without clinician-assigned labels.

Table 2: Verdict percentages for all $k = 10$ clusters ($N = 500$), identified by numeric ID only.

Cluster ID	Aligned (%)	Unaligned (%)	Uncertain (%)
0	4.25	20.20	75.55
1	21.08	42.02	36.90
2	5.15	22.29	72.57
3	0.88	23.51	75.61
4	10.00	4.00	86.00
5	0.15	28.38	71.47
6	4.60	3.45	91.95
7	0.56	28.65	70.79
8	4.38	36.25	59.37
9	49.23	3.46	47.31

5.4 Evaluator Availability Analysis

Finally, three authors evaluated a stratified sample of 25 concept-report pairs (8 Aligned, 8 Unaligned, 9 Uncertain) with MedGemma hidden, achieving 72% unanimous agreement.

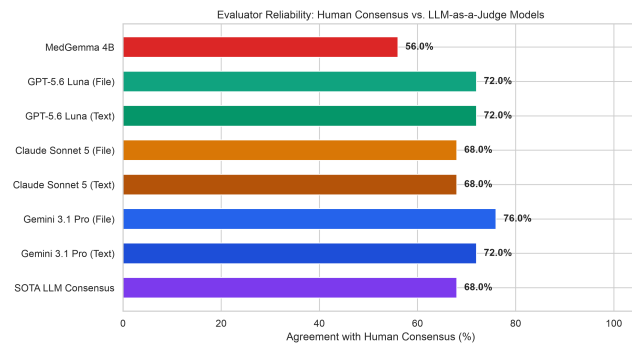


Figure 1: Evaluator agreement (%) with author majority consensus on 25 pairs.

Figure 1 and Table 3 compare MedGemma with three commercial LLMs (GPT-5.6 Luna, Claude Sonnet 5, and Gemini 3.1 Pro) using the authors’ majority verdict as the reference for 25 concept-report pairs. The commercial models receive the same evaluation material in two formats, file attachments and pasted text, to assess sensitivity to input presentation. MedGemma agrees with the reference on 56.0% of pairs, compared with 68.0%-76.0% for the commercial models. GPT-5.6 Luna and Claude Sonnet 5 achieve unchanged agreement across formats, while Gemini 3.1 Pro decreases from 76.0% with file input to 72.0% with pasted text. These results suggest higher agreement for the commercial evaluators, while a larger sample would help establish how consistently this advantage holds. We leave this broader evaluation to future research.

Table 3: Evaluator agreement (%) with human majority vote on 25 pairs.

Evaluator Model	File Format (%)	Text Format (%)
MedGemma 1.5 4B (Pipeline Judge)	-	56.0
GPT-5.6 Luna	72.0	72.0
Claude Sonnet 5 (High Effort)	68.0	68.0
Gemini 3.1 Pro	76.0	72.0

6 CONCLUSIONS

In this work, we presented an unsupervised concept discovery pipeline for 2D chest radiographs, adapting MedConcept through sparse autoencoders and report-based evaluation with MedGemma. The SAE achieves high embedding reconstruction fidelity ($MSE = 3.32 \times 10^{-5}$) with an average of 44 active units per image. Across 500 images and 12,996 concept-report pairs, mean per-image verdict percentages are 7.21% Aligned and 70.30% Uncertain. The predominance of Uncertain verdicts may partly reflect reporting omissions, since radiologists may leave normal anatomy or other visible features undocumented. Concept quality may also influence these results: noisy, redundant, or overly specific dictionary entries can complicate comparison with clinical reports.

Despite these textual limitations, hierarchical clustering reveals that visual alignment varies sharply by concept cluster rather than uniformly: the highest-scoring cluster reaches 49.23% Aligned and the lowest only 0.15%, a gap that remains stable across every clustering resolution we tested (Section 5.3). The audit of 25 concept-report pairs also highlights the role of evaluator reliability. MedGemma agrees with author majority consensus on 56.0% of pairs, compared with 68.0%-76.0% for the commercial models. These results suggest opportunities to improve automated evaluation, while a larger sample would help establish how consistently the observed differences hold.

Future research could refine the concept dictionary and validate feature-to-concept assignments to examine how these choices affect alignment. We also leave evaluation on a larger audit sample to future work. Expert review of the images could further assess whether the extracted concepts correspond to visible findings and influence model behavior.

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